

ECF	Error carried forward	SF	Significant figures error	M0	No A marks awarded
AE	Arithmetic error	BOD	Benefit of doubt	^	More is needed in answer
POT	Power of ten error	CON	Contradictory response	XP	Wrong physics
TE	Transcription error	IR	Irrelevant (part) response		

1a	$k = \frac{F}{e}$ $= \frac{0.180 \times 9.81}{0.036}$ $= 49 \text{ N m}^{-1}$	A	C1 A1
1b	$k = \frac{F}{e}$ $\frac{\Delta k}{k} = \frac{\Delta F}{F} + \frac{\Delta e}{e} = \frac{2}{36} + 0.02$ $\Delta k = \left(\frac{2}{36} + 0.02\right) \times 49$ $= 3.7$ $= 4 \text{ N m}^{-1}$ $\text{Accept } \Delta k = \frac{k_{\max} - k_{\min}}{2}$	A	C1 A1
1ci	F is weight of the column of liquid above the area A $\rho = \frac{F}{A} = \frac{m_{\text{fluid}} g}{A}$ $= \frac{(V_{\text{fluid}} \rho_{\text{fluid}}) g}{A}$ $= \frac{(hA \rho_{\text{fluid}}) g}{A}$ $= h \rho g$	A	B1 B1 A0
1cii	$U + ke = mg$ $U = mg - ke$ $= (0.180)(9.81) - 49(0.030)$ $= 0.2958 \text{ N}$ $\Delta p A = \Delta h A \rho g$ $U = V \rho g$ $0.2958 = (2.0 \times 10^{-5})(9.81) \rho$ $\rho = 1500 \text{ kg m}^{-3}$	A	C1 C1 A1